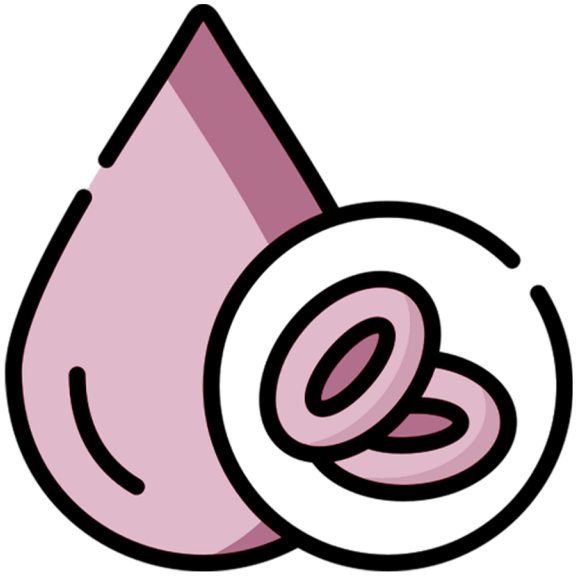


MICROTUTORIALS

BLOOD MICROBIOLOGY PRACTICAL 2025-2026

DR. MUHAMMAD ASHRAF

Scan to join group

Pulse.

Blood culture

Indications:

- **Fever of unknown origin**: Fever of ≥38.3ºc lasting for more than 3 weeks with failure to reach diagnosis

- **Suspected bacteremia**

  - Fever/hypothermia
  - Tachycardia
  - Confusion
  - Severe hypotension
  - Elevated inflammatory markers

- Infective endocarditis - Brucellosis, typhoid

- Localized infection with systemic course: osteomylitis ,meningitis

steps

- **1- Blood collection and sampling**
- **2-Blood Culture**

Blood collection

- **Two or more blood samples taken through venipuncture**



- **Drawn from different veins.**

- **During the fever attack.**

Under Aseptic technique

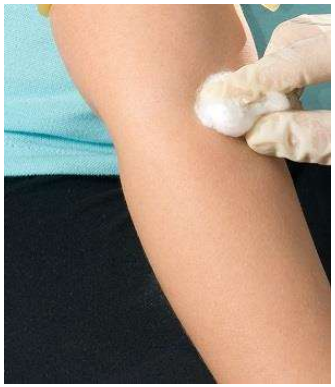
aseptic technique before venipuncture



- **1- Hand washing**



- **2-Wearing gloves**

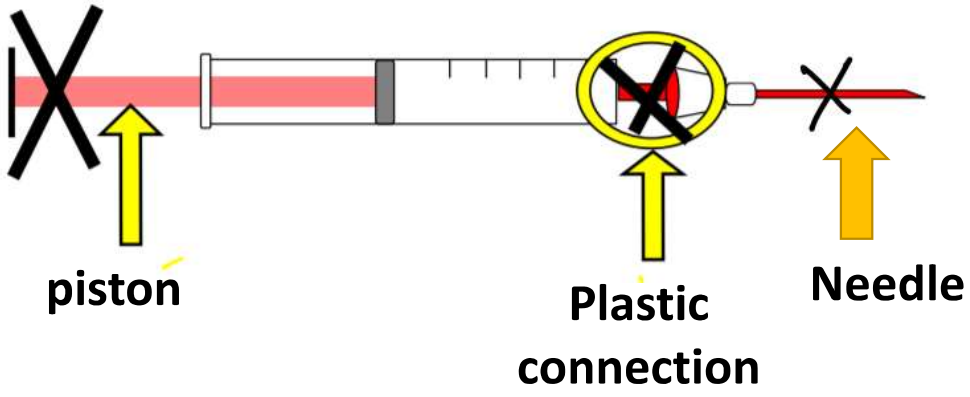


- **3- Apply skin antisepsis (alcohol 70%) in circular movement**

Important tips

- 1- **NEVER** touch the skin or palpate the vein after applying antiseptic

- 2- **NEVER** touch the sterile areas of the syringe

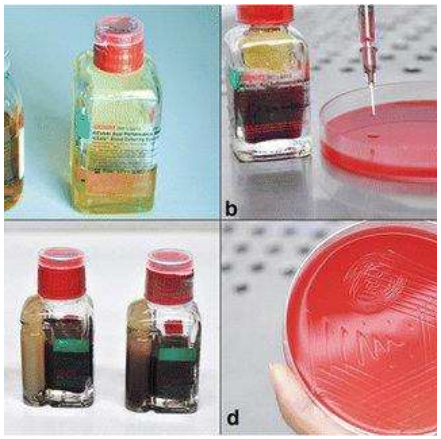


To avoid Contamination , contamination = false result

Types of blood culture



- **Automated**



- **Manual**

Manual Blood culture

- **1- Conventional Traditional blood Culture**



- **2- Diphasic “Castaneda”**



Conventional Traditional blood Culture

- **Blood sample**



- **Fluid media "Broth"**

  1-brain-heart infusion broth





  2-tryptone soya broth



Blood to fluid media ratio should be **1**:**10** (**1ml blood** : **10ml fluid media**)
e.g. (**5ml blood sample** : **50ml fluid**)
(**10ml blood sample** : **100ml fluid media**)

Why we must add blood sample to fluid media?

1- nutrition for the organism to grow

2-Dilution of patient's "Host's" antimicrobial factors

- **Antibodies**

- **complement**



Subculture

Under Aseptic technique



Nutrient agar



Blood agar



Chocolate agar



MacConkey's agar

When to subculture:

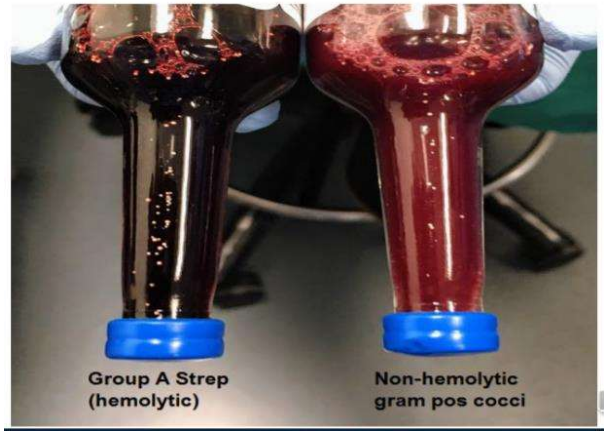
- Every Other day (48 hours)

or

- **when changes in broth bottle are observed**



Turbidity



Hemolysis

Advantages: Low cost

Disadvantages: High risk of contamination due to repeated subcultures

Castaneda medium (Diphasic culture media)

Broth Fluid bottle + **solid agar slope**

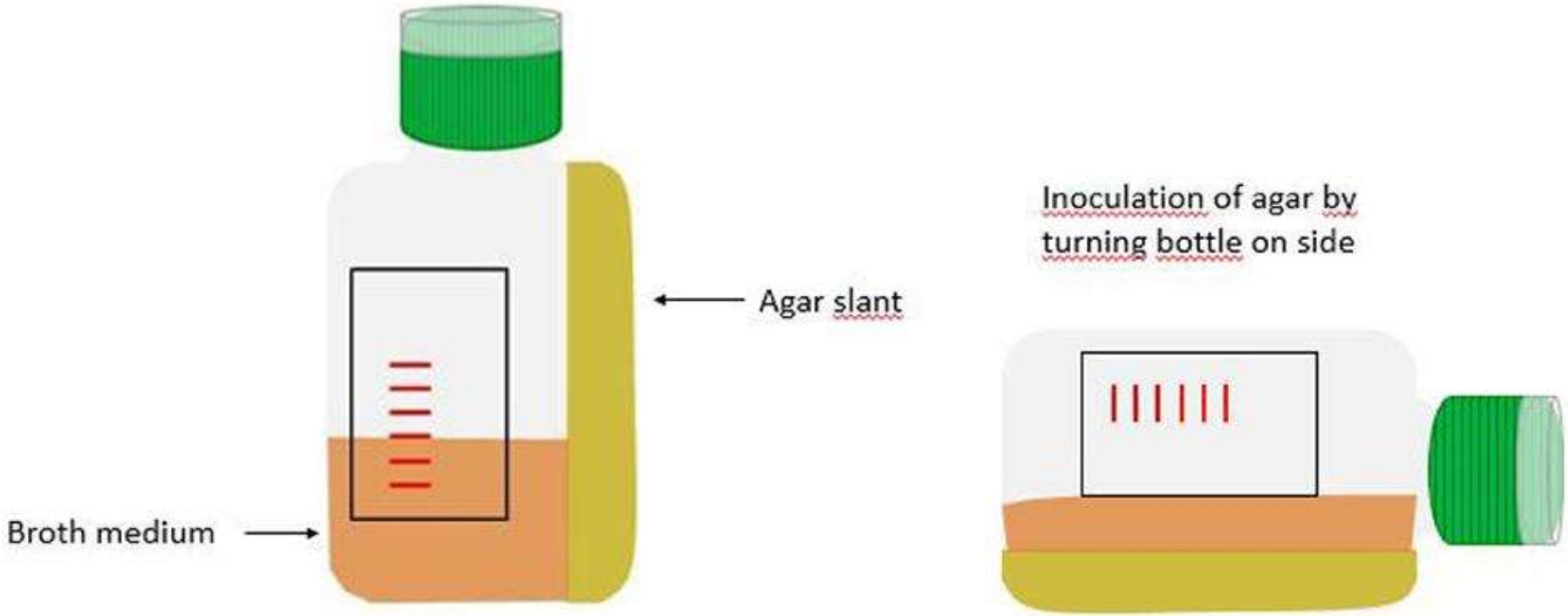
Agar slope is examined Twice daily

Advantages:

No need for subculture

reduces the possibility of contamination




Agar slant
Broth medium
Inoculation of agar by
turning bottle on side

Incubation period

in normal samples

- **1 week in 37 C**

If Brucellosis is suspected

- **3 weeks into Tryptic soya diphasic medium**



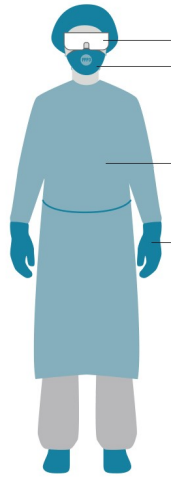
Brucellosis

- **Brucellae are highly infectious organisms**
- If brucellosis is suspected:

all the diagnostic tests, should be done as the following:

- within special lab
- biosafety cabinet level 3
- with appropriate personal protective equipment.





Diagnosis of Brucellosis

- **Acute Brucellosis**



Direct method

Culture on special media

- Microscopic examination
- Biochemical reactions

- **Chronic Brucellosis**



serology

The most commenly used method in diagnosis of brucellosis

Serology "Serum Agglutination test"

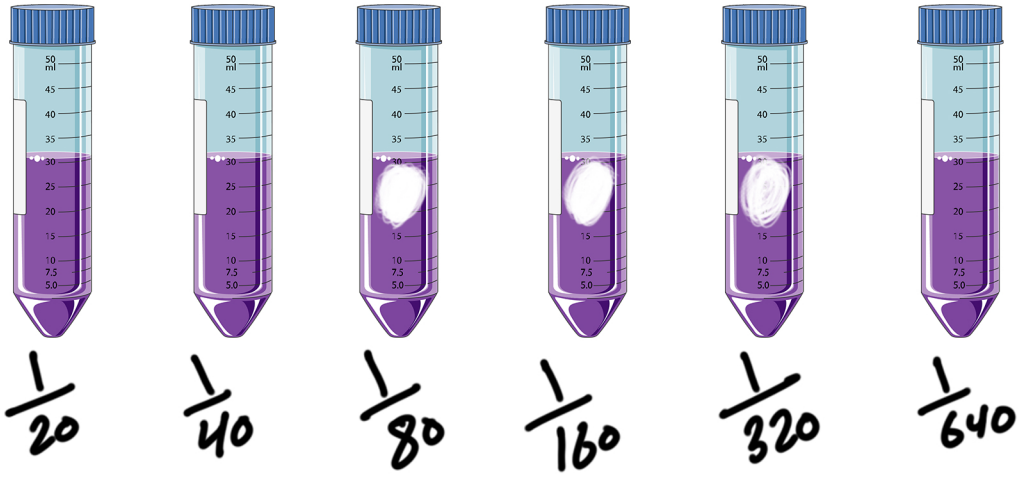
Most common used method in diagnosis of brucellosis

Used in cases of chronic brucellosis

* SAT: Serum Agglutination test

"Titer > 1/80 is diagnostic"

NB: In SAT, Prozone phenomenon may occur due to High Ab/Ag ratio



How to overcome:

1- Coombs ragent

2- Using more tubes (higher range of dilution)

Organisms Can be found in blood culture causing bacteremia & septicemia

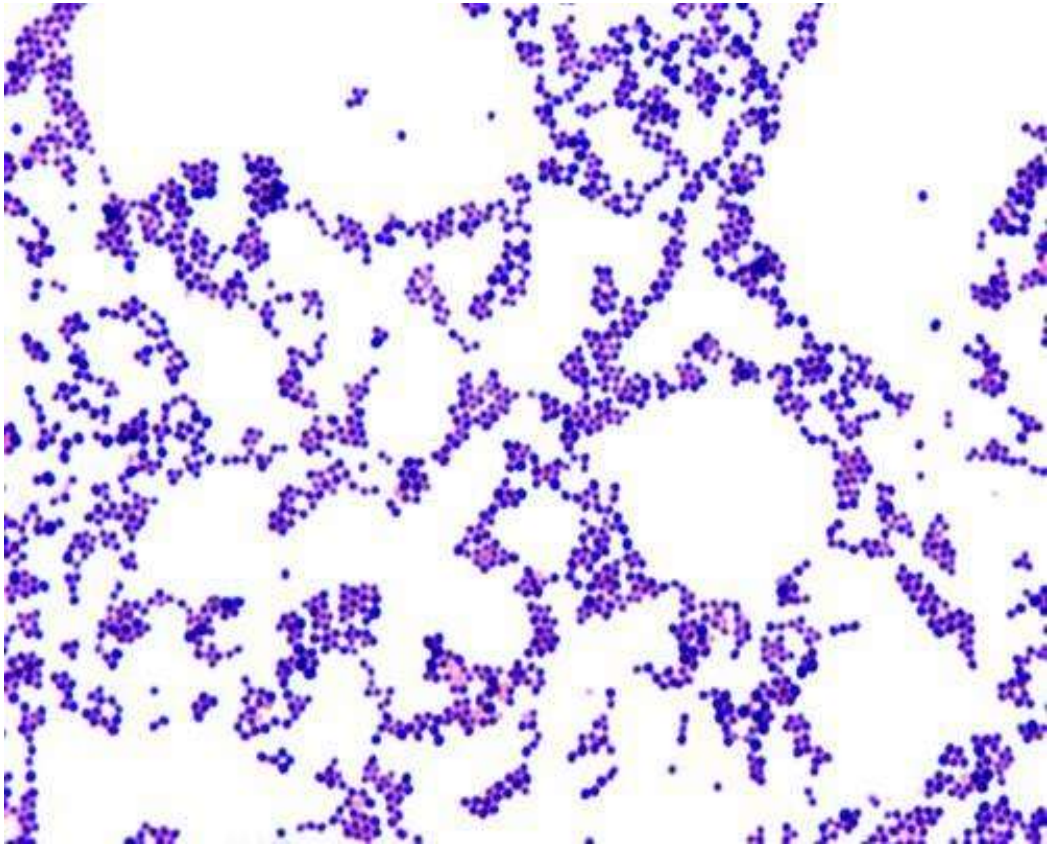
- **Staph Aureus**
- **Strept. Pyogenes**
- **Neisseria meningitidis**
- Members of Enterobacteriaceae :

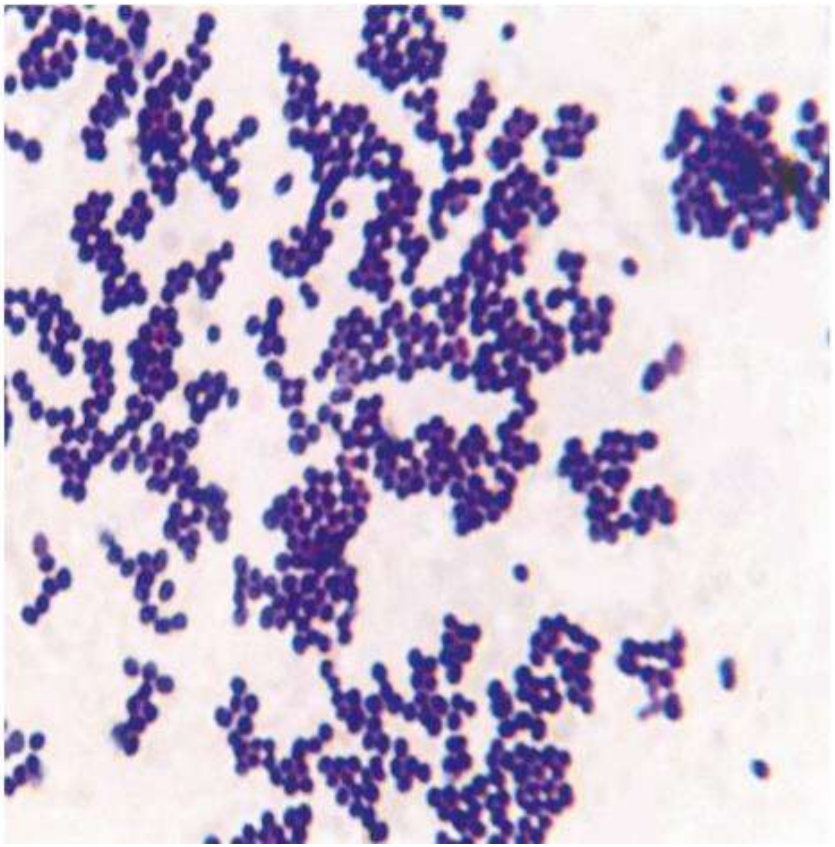
  1- E. coli

  2- Klebsiella

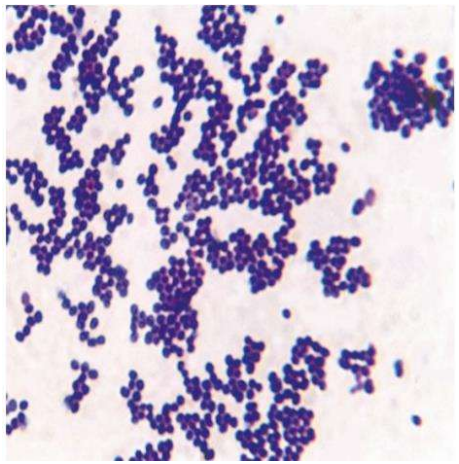
  3- Salmonella
- **Pseudomonas aeruginosa**
- **Candida albicans** ≈ fungoi

Staph Aureus

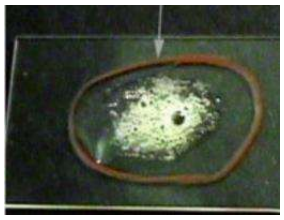


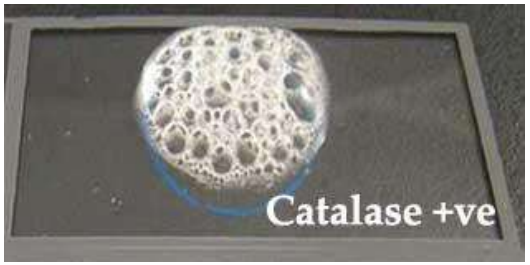


Staph Aureus

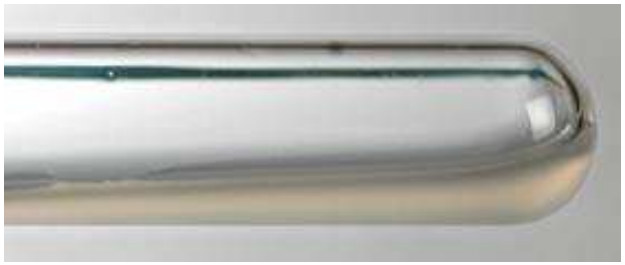


Gram positive cocci arranged in grape like clusters





catalase +ve



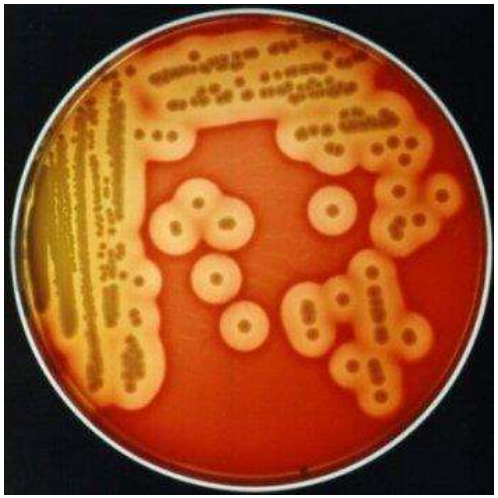


Coagulase +ve

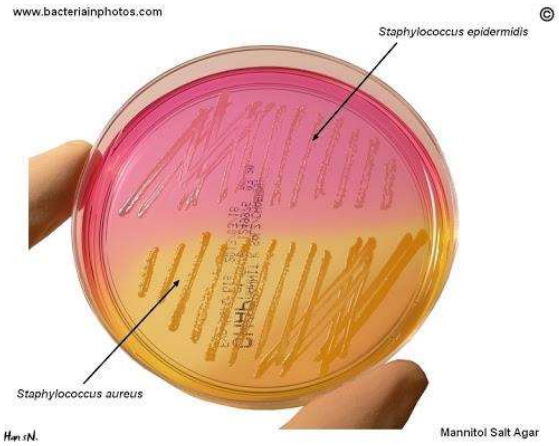
Staph Aureus



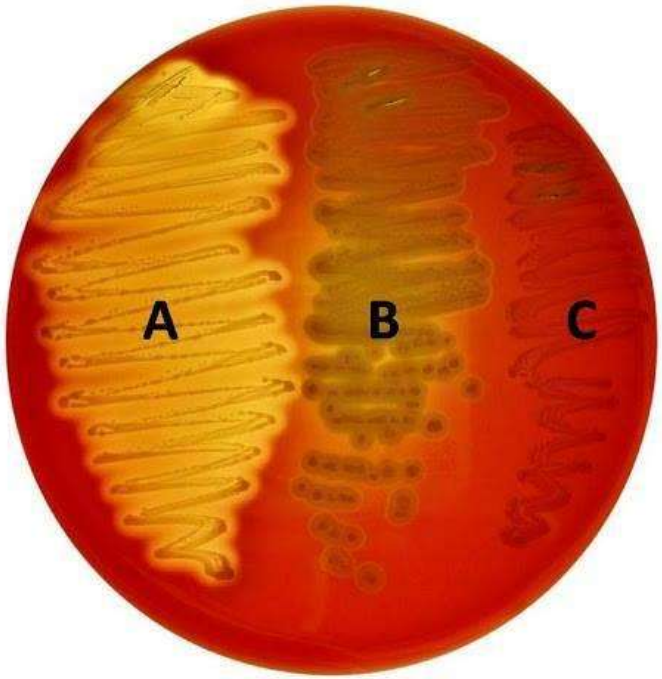
Golden yellow endopigment production



complete β hemolysis on blood agar

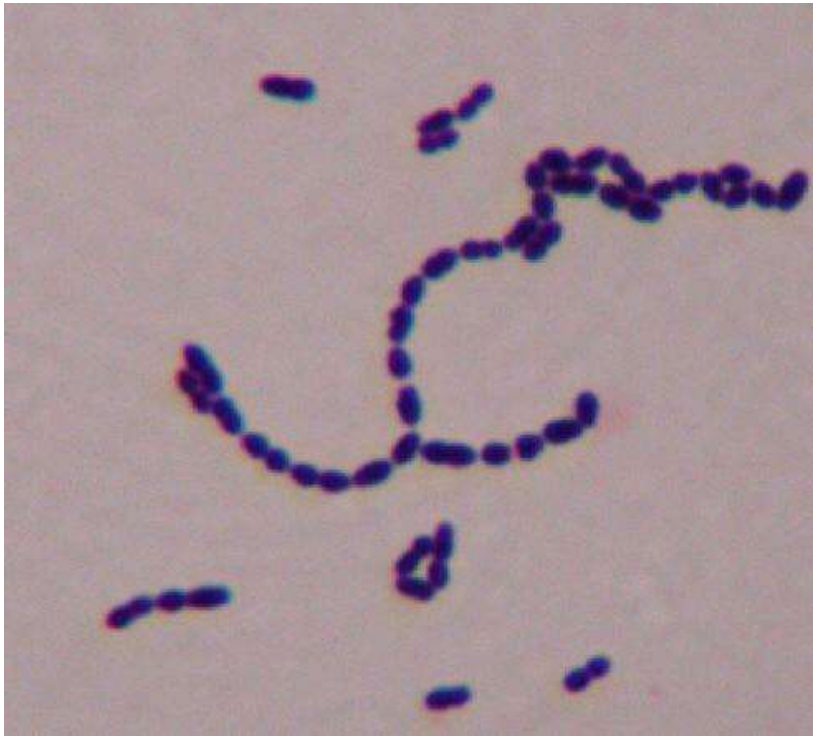


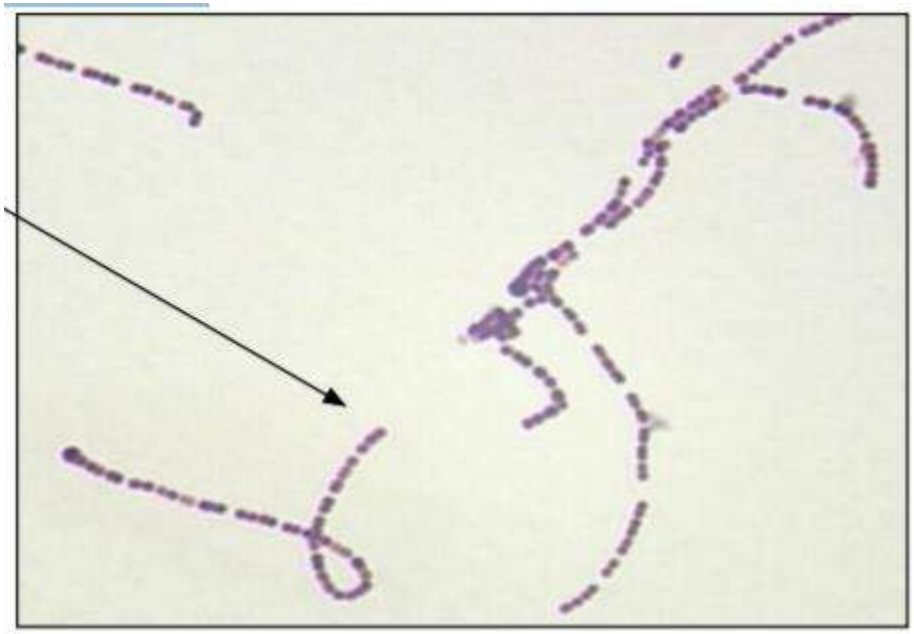
Yellow colonies on mannitol salt agar (mannitol fermentation)



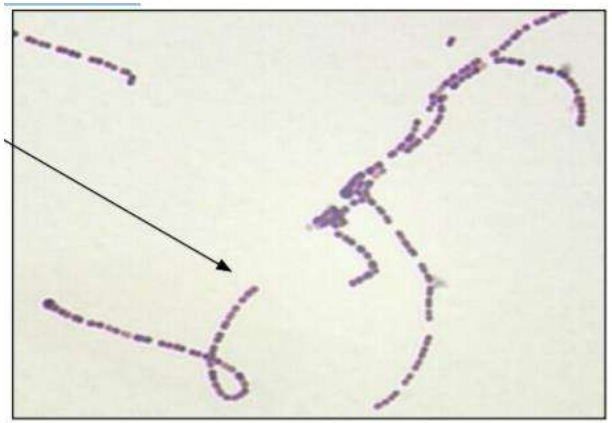
Copyright © 2006 Pearson Education, Inc., publishing as Benjamin Cummings.

Strept. Pyogenes

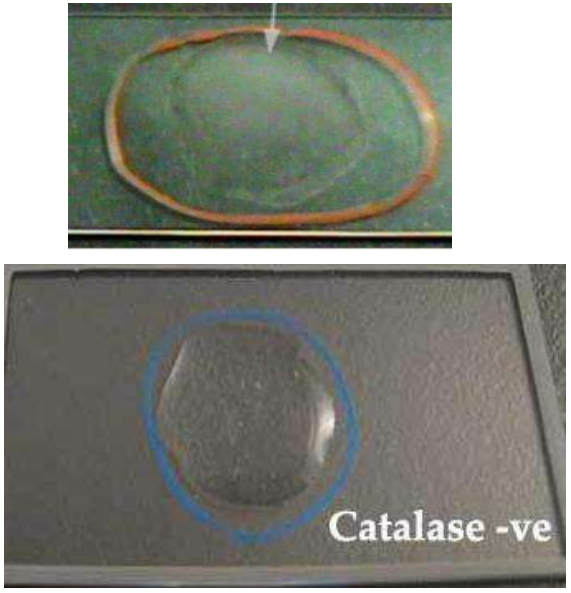




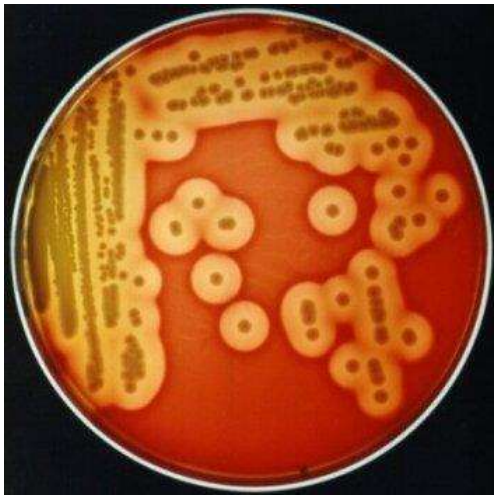
Strept. Pyogenes



Gram positive cocci arranged in chains

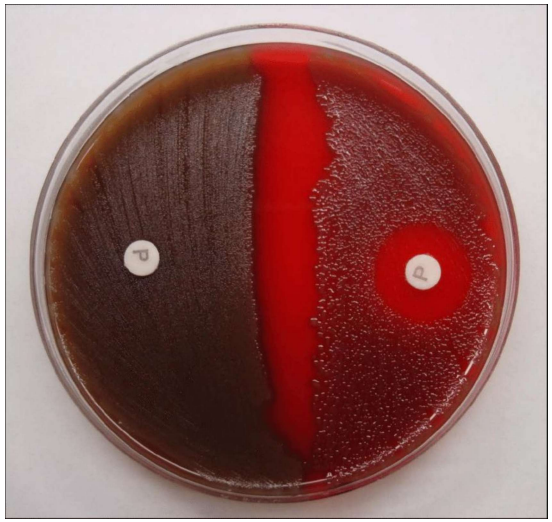


catalase -ve



complete β hemolysis on blood agar

Strept. Pyogenes



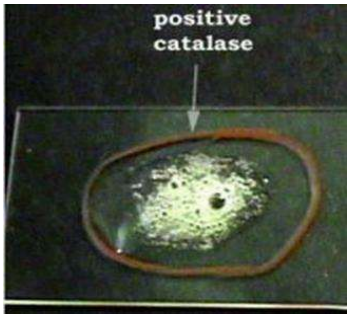
Sensitive to bacitracin

•**Catalase test:**

To differentiates *Staphylococci* (catalase positive) from *Streptococci* (catalase negative). as they are both gram positive cocci under the microscope.

Principal of catalase test:

$2H_2O_2 \quad H_2O + O_2$



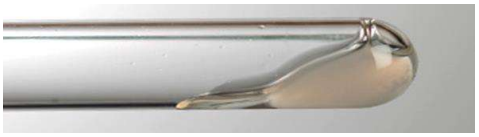
This is done by picking the colony on a clean thin glass rod and immersing it in a few drops of hydrogen peroxide. Rapid effervescence indicates gas production and a positive test

The result of this test is **Positive**.

➢ **Coagulase test:**

To differentiate between ***Staphylococcus aureus*** **(the most pathogenic) and coagulase negative *Staph* (*epidermidis and saprophyticus*)**

Coagulase causes **clotting of citrated or oxalated human or rabbit plasma through converting fibrinogen into fibrin.**



It can be detected by:

•**Slide method:** This detects bound coagulase.

Coagulase positive Staphylococci clump within 15 seconds because coagulase precipitates the fibrin in the plasma on the cell surface.

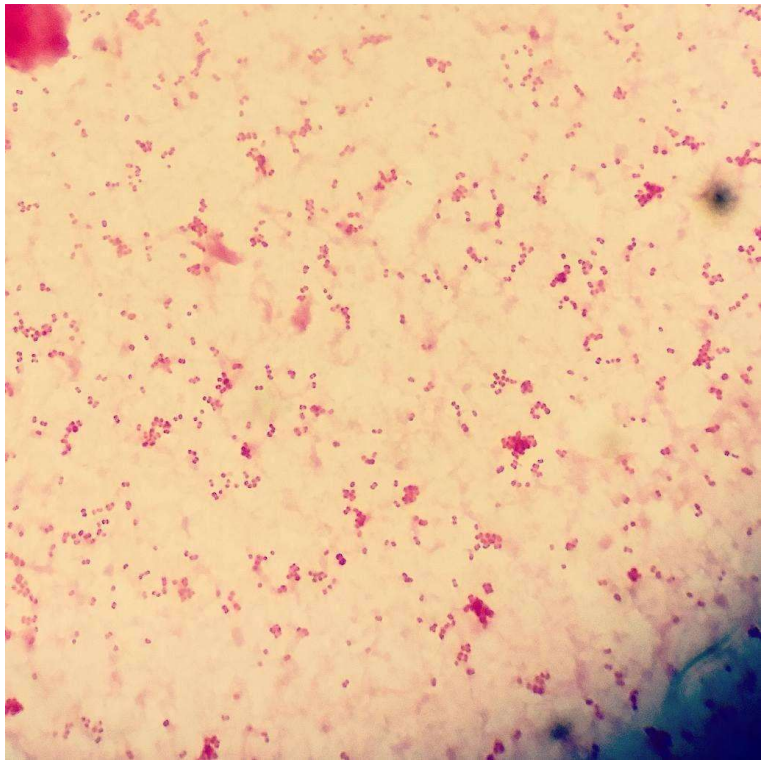
•**Tube method:** This detects free coagulase.

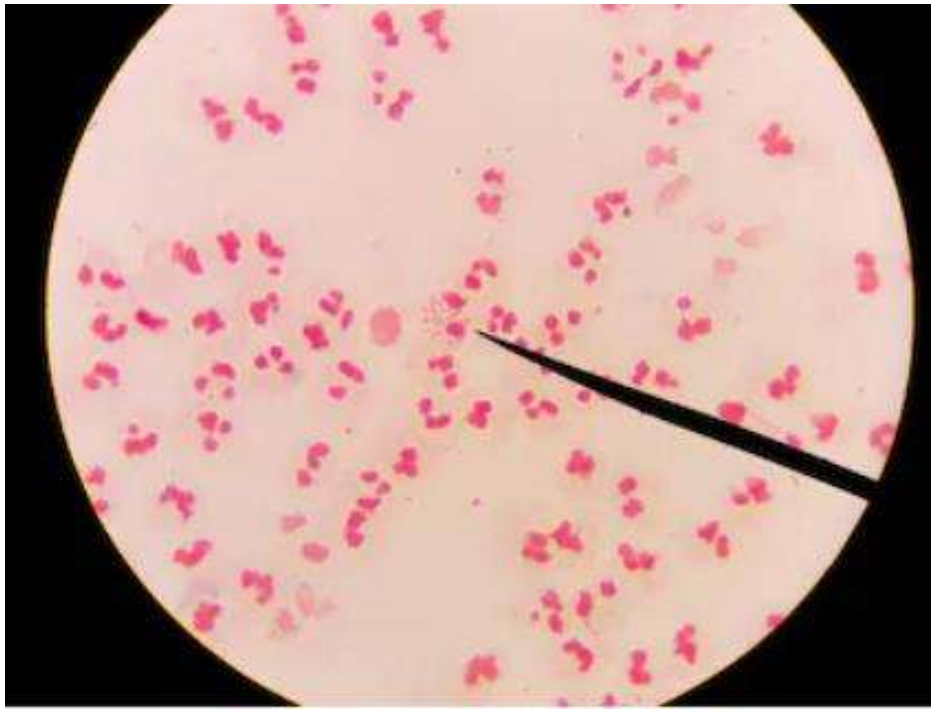
•Test tube contains plasma inoculated with a staphylococcal colony.

•A control tube containing 0.5 ml of the used plasma.

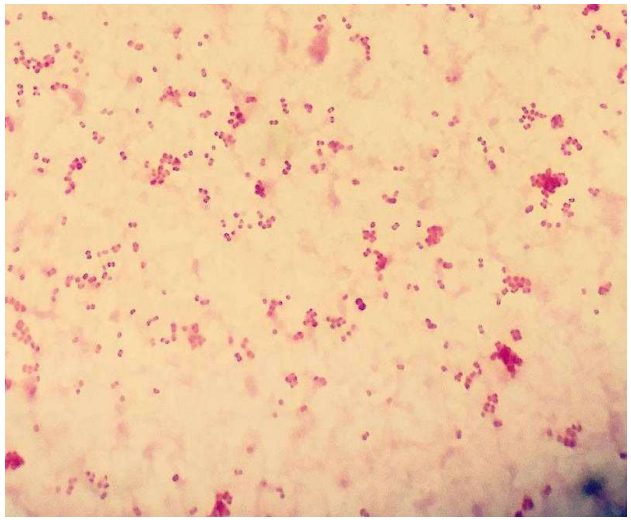
•Both tubes are incubated at 37ºC and inspected hourly for 6 hours to detect clot formation. The result of this test is **Positive**

Neisseria meningitidis

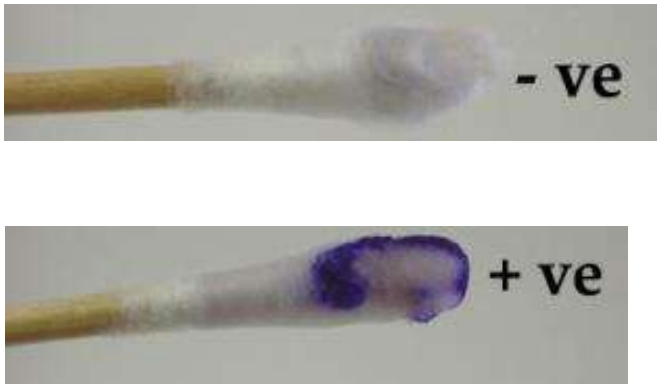




Neisseria meningitidis



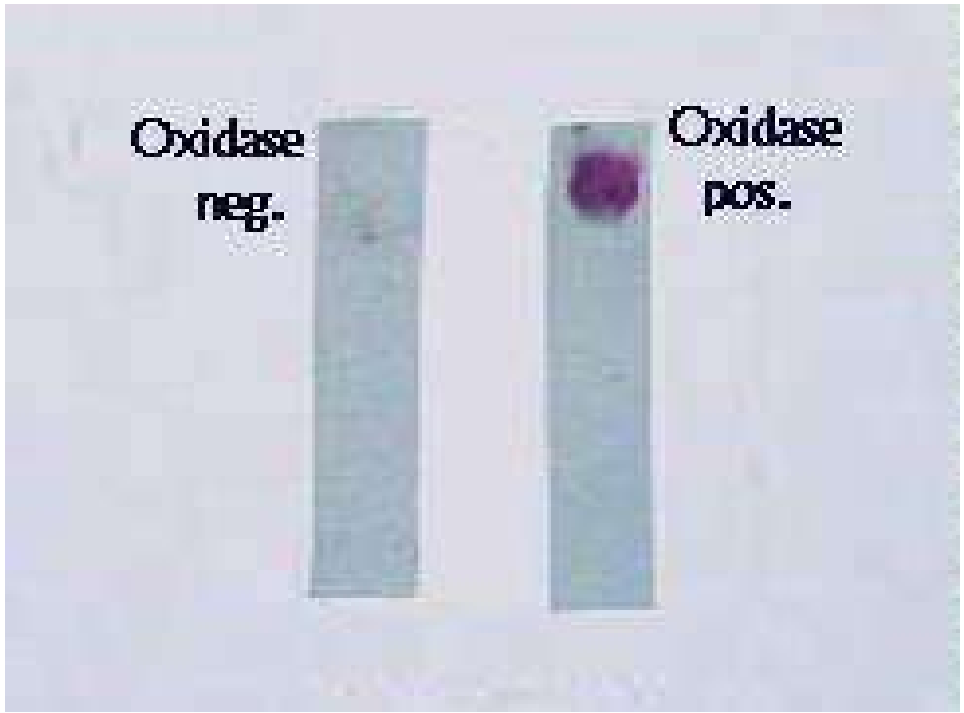
Gram -ve diplococci arranged in pairs



oxidase +ve



Isolated from blood on chocolate agar

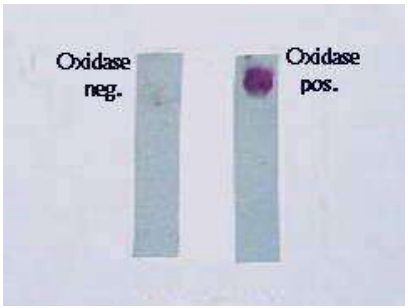

Oxidase
neg.
Oxidase
pos.

Enterobacteriaceae



Gram negative bacilli with no special arrangement

- **Gram negative bacilli , Non- spore forming, no special arrangement**





Oxidase negative

Reduce nitrates to nitrites



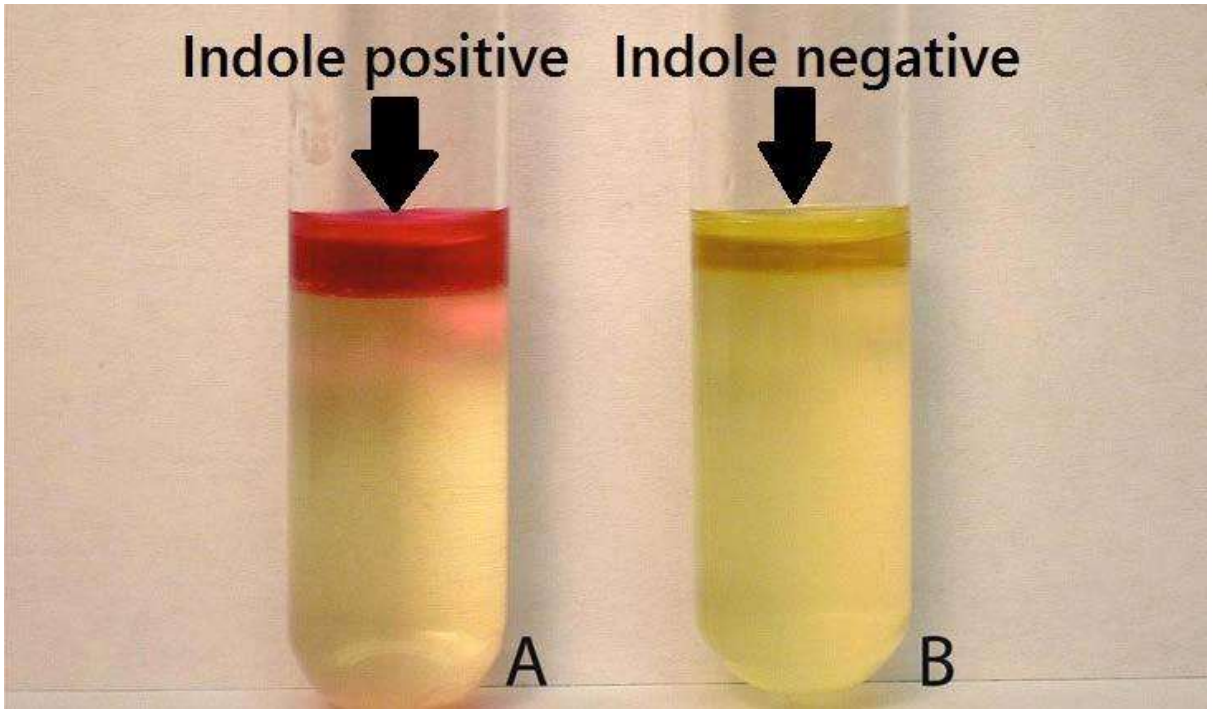
Grow on MacConkey's agar

Important tests



Important tests





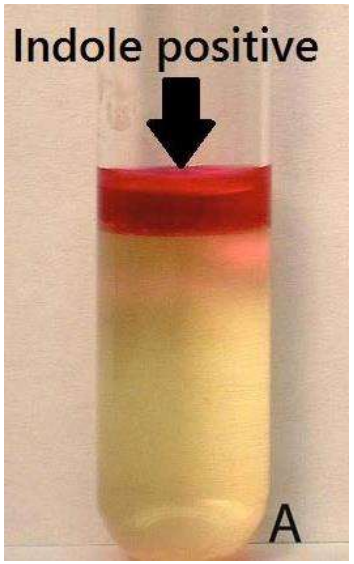


E. Coli



Lactose fermenter on MacConkey's agar





indole +ve



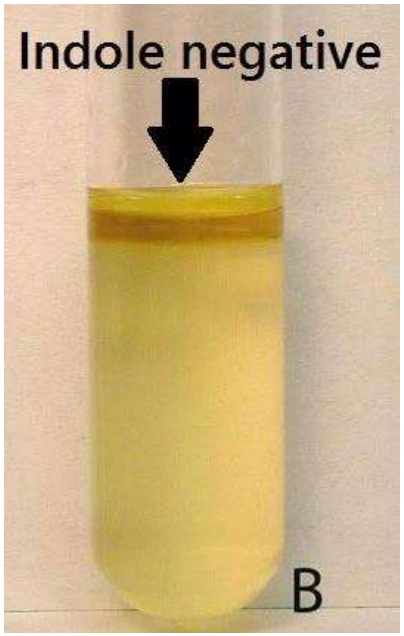


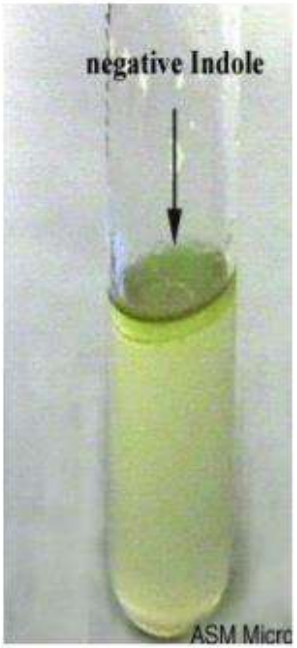
Ferment 3 sugars on TSI agar with acid and gas production

Klebsiella



Lactose fermenter on MacConkey's agar





indole -ve





Ferment 3 sugars on TSI agar with acid and gas production

Salmonella



Non-lactose fermenter on MacConkey's agar

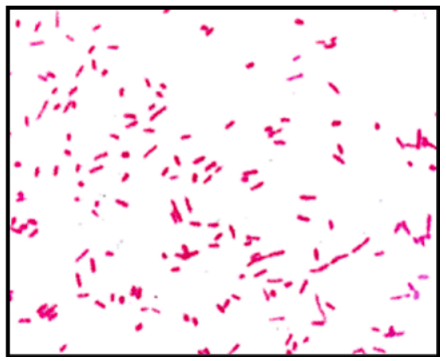


Ferments glucose without gas production



Some species are H2S positive

Pseudomonas aeruginosa



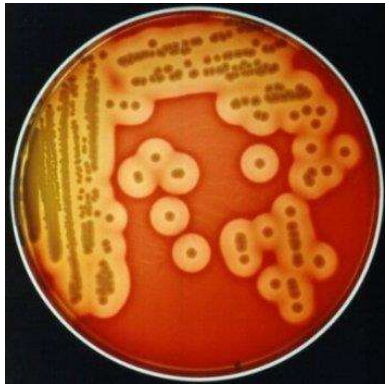
Gram negative bacilli with no special arrangement



Green Exopigment on nutrient agar



Non-lactose fermenter on MacConkey's agar

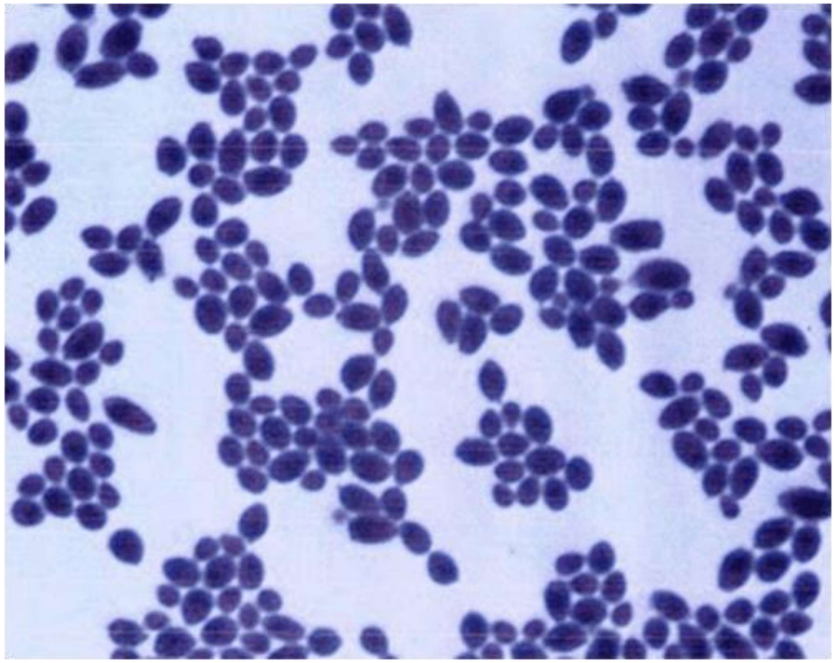


complete β hemolysis on blood agar

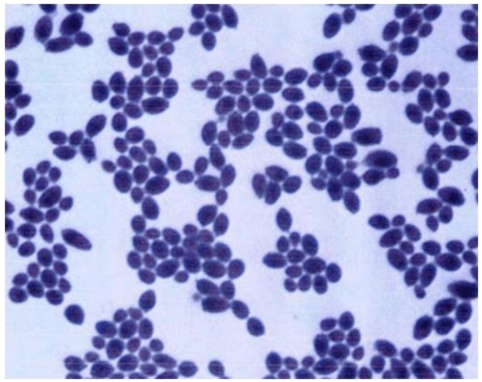


oxidase +ve

Candida Albicans



Candida Albicans



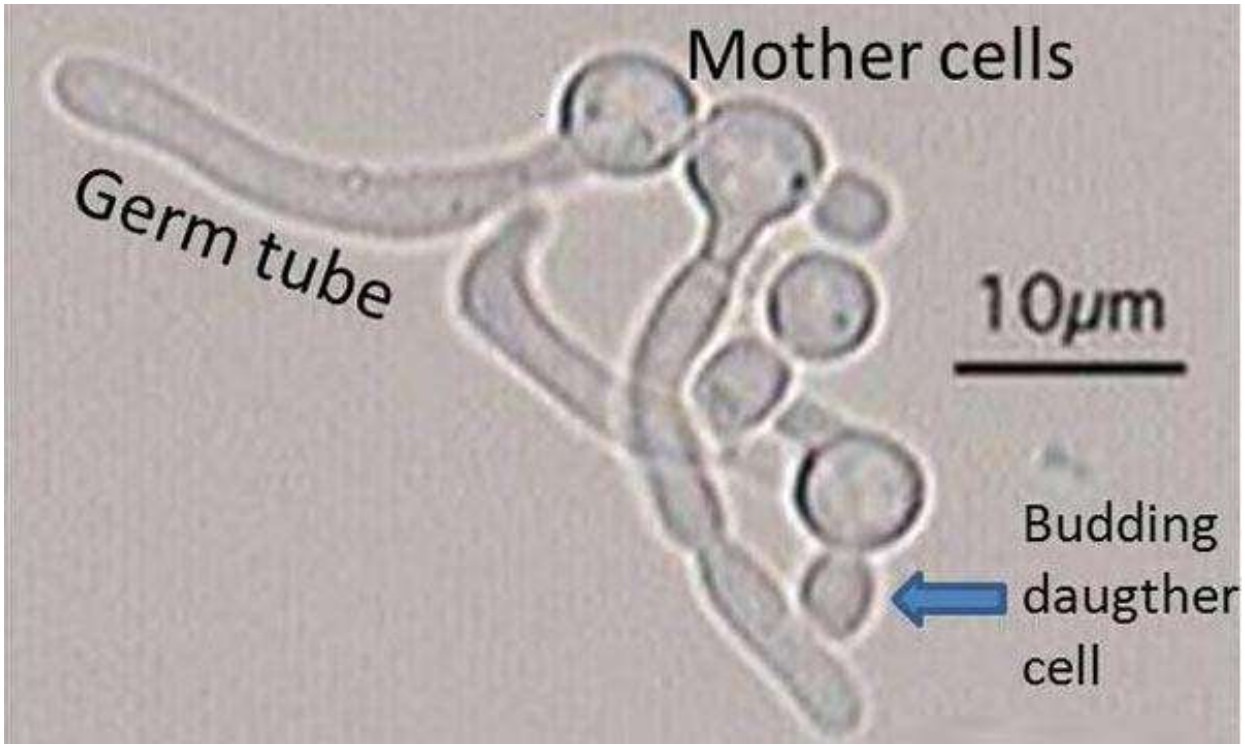
Gram positive oval cells" yeast like"



Creamy colonies on Sabouraud's dextrose agar



+ve germ tube test

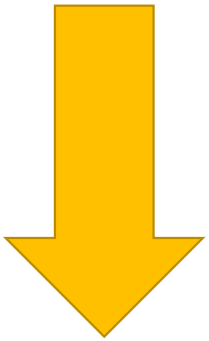


Germ tube test. Yeast cells of Candida albicans begin to form true hyphae after serum inoculation for 3 hours at 37˚c.

Interpretation of blood culture results:

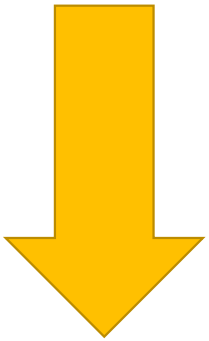
- **2 or more positive results for the same organisms**
- **2 negative results**
- **One positive result and one negative**

2 or more positive results for the same organisms



Person has a blood infection with that microbe

2 negative results



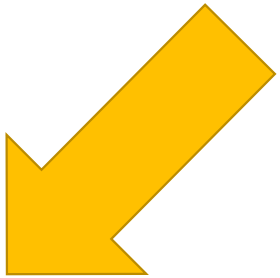
- **There is no infection**

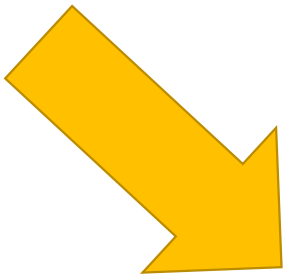
- **Some microbes are difficult to grow in culture**

- **localized infection e.g. deep seated abscess**

- **Virus**

One positive result and one negative result





If the positive result is for commensal "flora"

If the positive result is for pathogenic organism





Contamination "False result"

Person has a blood infection with that organism